ThermoTwin-IN

A Personal Heat-Strain Digital Twin for Indian Outdoor Workers

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- Educational proof of concept using synthetic data

Problem and motivation

Outdoor workers can experience changing heat exposure and physical strain

A generic threshold may miss differences from a person's baseline

- Focus: combine personal context and dynamic sensor/environment readings

Two data streams

Personal/historical: age, work context, baseline heart rate, sleep and hydration context

Dynamic: temperature, humidity, activity, current heart rate, exposure duration and sensor quality

- No real worker data is used

Digital-twin workflow

Personal profile + dynamic stream → state fusion → synthetic ML score

- Compare current scenario with a hypothetical rest/shade scenario
- Show warnings when sensor quality is delayed or noisy

Implementation

Python, Pandas, NumPy, scikit-learn Random Forest and Streamlit

- CSV dataset with 2,000 synthetic records
- Reproducible train/test split and model-training script
- Interactive inputs and side-by-side scenario dashboard

Demo plan

1. Set a worker profile and environmental conditions
2. Adjust temperature, activity, heart rate and exposure
3. Observe personal-baseline deviation and score
4. Compare a hypothetical rest/shade scenario

Safety, limits and trust

- All data and labels are synthetic
- Scenario assumptions are illustrative, not validated physiology
- Not for diagnosis, treatment or real-world safety decisions
- Real use requires validated metrics, external validation and expert oversight

Next steps

Use suitable ethically approved data and validated heat metrics

Evaluate uncertainty, bias, missing-data behavior and generalization

Consult workers and occupational-health experts

Repository and demo-video links: add before submission

Conclusion

ThermoTwin-IN demonstrates a personal, changing virtual worker state

It integrates two data streams and compares hypothetical scenarios

- Focus: transparency, data-quality warnings and safety-conscious design